

Flowing-Haptic Sleeve: Research on Apparent Tactile Motion Applied to Simulating the Feeling of Flow on the Arm

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ABSTRACT

Human-computer interaction, VR and other fields have raised higher requirements on haptic feedback. Due to good expressiveness, low cost and low technical complexity, the current researchers focused on applying the principle of apparent tactile motion to generate linear or pattern tactile motion feedback. However, there still exist limitations because these applications cannot restore the high-fidelity tactile experience that can characterize the tactile texture and fail to take the matching of real application scenarios into account. **Flowing-Haptic-Sleeve** was developed to combine apparent tactile motion and tactile texture to generate a situation-based high-fidelity sense of flow on the arms. This research adopted a multi-factor experiment design and conducted psychophysical experiments on tactile stimulus with 11 participants to explore the feasibility and control space of generating such tactile motion feedback. The results are useful to complement current researches on apparent tactile motion by integrating tactile motion with tactile texture.

CCS CONCEPTS

• Human-centered computing → Sound-based input/output;
• Haptic devices; • Laboratory experiments;

KEYWORDS

Tactile devices; Haptic illusions; Haptic experience design; Virtual and augmented reality

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1 INTRODUCTION

Realistic and real-time haptic feedback has the potential to significantly enhance the accuracy [6] and enjoyment of Human-Computer Interaction. This technology is widely used and low-cost in the fields of Virtual Reality, but it mostly only provides simple haptic feedback like alerting or shaking (e.g., eccentric-mass motor [24]). The homogenous haptic feedback and its own weak information carrying capacity make it impossible to meet the demand in the VR fields for immersive experience.

In order to explore the possibility of more abundant and realistic haptic feedback [21, 23], current researches and applications choose voice coil tactor array for tactile motion feedback to output linear [3] or pattern [11]. However, there still exist limitations and development potential in the current application of the apparent tactile motion.

In this study, the TECHTILE toolkit [15] and Stereohaptic [12] framework were referred to design the Flowing-Haptic Sleeve. The initial conception of this apparatus shown in Figure 1 was based on the lateral line system of fish [8]. It aimed to produce a dynamic tactile motion combined with tactile texture on the arm to restore the realistic sense of flow. In addition, parametric modeling of the sense of flow was carried out, followed by the related research on psychophysical experiments and control space. This research mainly responds to three issues:

- Can the tactile motion stimulus generated by the combination of apparent tactile motion and tactile texture be clearly perceived and recognized after parameters control?
- To estimate the parameters of the sense of flow and control space based on linear motion.
- Can a unidirectional linear actuators array produce a sense of steering through parameter control?

2 BACKGROUND

Sequential stimulations at two locations on the skin create a perception of illusory moving sensation between the two points [10, 11]. A typical vibrotactile duration of stimulus is defined by a frequency and amplitude and also characterized by stimulus onset asynchrony (SOA), onset and decay [12]. While Kirman and Neuhaus's study [9, 16] showed that SOA and duration play a significant role in generating the apparent tactile motion, Israr's study in [11] and Zhao's study in [24] responded the conclusions from previous researches that the frequency and the amplitude of tactile stimulus had little influence on the effectiveness of apparent tactile motion, and simultaneously showed the best-fit function relating optimal SOA to duration is: $SOA = 0.32 \times DURATION + 47.3$ [11]. However, many studies preferred to use various fixed-frequency sine wave

signals instead of texture vibrations signals [5] sampled from real objects as tactile signals. Therefore, such dynamic tactile stimulus is difficult to produce high-fidelity experience.

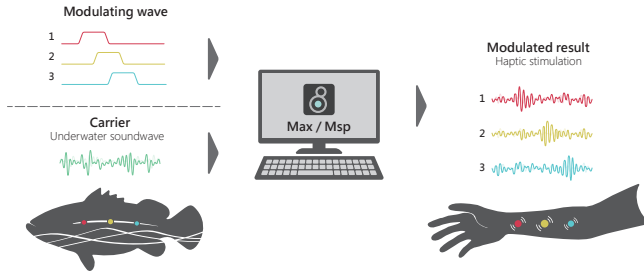


Figure 1: Concept Illustration for the generation of tactile signal

According to Manfredi's study in [14], during active or passive touch, the skin surface rubs against the surface of the object to generate complex high-frequency vibrations, which carry information about the surface texture of manipulated objects. Briefly, these complex high-frequency vibrations largely determine "Texture", and the frequency composition of elicited vibrations is texture-specific and highly repeatable [2, 4, 20]. We believed that the current research and application of tactile texture and apparent tactile motion had the potential to complement each other, to generate dynamic high-fidelity tactile motion feedback on the skin with low-resolution grid of vibrating actuators [13]. Amplitude-Modulation means "The amplitude of the carrier wave varies in accordance with the instantaneous amplitude of the modulating wave" [7]. When it comes to tactile motion stimulus, the vibrotactile stimulus (e.g., fixed-frequency or tactile texture signal) can be regarded as a carrier wave.

The system generates a modulating wave to perform amplitude modulation on the carrier wave. [24]'s results showed that location of illusory sensations between two vibrating points are well represented by modulating their amplitudes with a logarithmic function.

3 FLOWING-HAPTIC SLEEVE

In this section, the hardware and software architecture of the **Flowing-Haptic Sleeve** are introduced.

3.1 Description of the Flowing-Haptic Sleeve

The generation principle of tactile signals is shown in Figure 1. The output interface is composed of a 7.1-channel sound card (Figure 2a) connected to 3 sets of 2×15 W B Type audio amplifiers (Figure 2d), and the output of the tactile stimulus signals is transmitted through 3 pairs of voice coil tactors (Acoupe Laboratory Vp2 shown in Figure 2c) arranged linearly on both sides of the Flowing-Haptic Sleeve (Figure 2a). 3D printing technology was finally used to produce the exoskeleton structure. We made 3 pairs of voice coil tactors arranged linearly at a distance of 60 mm [1], and also partially designed the wearing comfort.

Figure 1 shows the low-pass filter of Max/Msp (<https://cycling74.com/>) is applied to filtrate above 1000 Hz frequency [22] of the carrier, followed by the amplitude modulation on the 6 groups

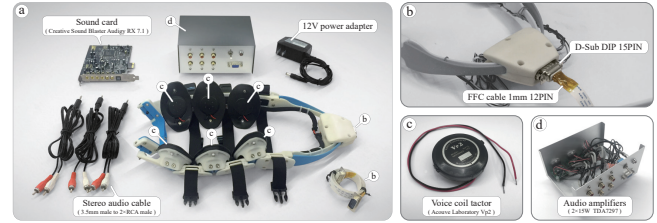


Figure 2: (a) All hardware for Flowing-Haptic sleeve. (b) The overall signal circuit is composed of common electronic components. (c) Actuators for the device. (d) The amplifier circuit and the connector are sealed to increase the stability of the device.

of audio signals. The basic modulating waveforms are shown in Figure 3. According to the previous research on apparent tactile motion in onset function, [24] the onset time of the modulating wave was set to 50 msec.

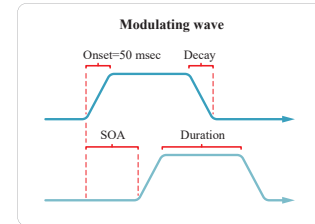


Figure 3: The components of the modulating wave.

3.2 Experiment Setup

This study had three stages of experiments. The first stage was to explore the recognition of the tactile stimulation, the second stage was to discuss the sense of flow generated by the tactile stimulation, and the third stage was to explore the underwater steering state of the tactile stimulation.

A total of 11 participants (six males; 24-28 years old, average = 24 years) were recruited in the three-stage experiment of this study, and all of them had not participated in the development of haptic technology and were in good health. The time interval for each stage of the experiment was more than one week, and all participants had signed a consent form before each experiment.

4 EXPERIMENT 1: RECOGNITION OF THE TACTILE STIMULATION

This experiment had two purposes based on the function relating optimal SOA to duration [11]. We focused on the two variables of duration of overall stimulus (DOS) and duration to estimate the optimum control space which allowed the participants to clearly feel the transmission of tactile stimulus. The satisfaction with the tactile experience was also estimated.

4.1 Methods

This experiment evaluated the difference in perception by changing two variables (Duration and Interval) of the modulating wave.

According to van Erp [22] and [17]'s study, we set 3 Duration levels, namely **Dur600** (600 msec), **Dur300** (300 msec) and **Dur150** (150 msec). In addition, the Interval levels of 4 different scales were selected, namely **Int0** (0 msec), **Int150** (150 msec), **Int300** (300 msec) and **Int600** (600 msec). All combinations of variables were shown in Supplemental File 1. In each trial, we used two different Likert Scales to measure the sensing scales of Recognition Rate and Satisfaction, and performed repeated measures ANOVA to analyze Recognition Rate and Satisfaction. (Supplemental File 2):

4.2 Results of Experiment 1

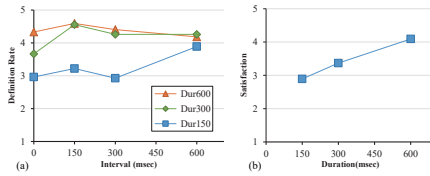


Figure 4: (a) Average recognition rate of the three durations at different interval levels. (b) Average satisfaction at different durations.

4.3 Discussion

The ANOVA results of Recognition Rate showed the main effect of Duration and Interval, and the Duration-Interval interaction was not significant. It could be inferred that the participant relied on intuitive decision to intuitively simplify and recognize the tactile signal as a specific rhythm when it was **Int150**. When the interval between DOS was **Int600**, there was enough reaction time for the participant to make a rational decision and consciously recognize the tactile signal, and thus more flaws were felt. With the improvement of duration, the participants could fully perceive and understand the transmission of tactile signals.

5 EXPERIMENT 2: SENSE OF FLOW GENERATED BY THE TACTILE STIMULATION

This experiment used **Int600** which had a higher level of Recognition Rate in the results of experiment 1 as the Interval between DOS. Then we had three purposes. Firstly, the impact of changing the carrier and modulating wave on the sense of continuous motion would be studied. Secondly, the sense of speed was to be evaluated. Thirdly, the satisfaction with the tactile experience would be evaluated.

5.1 Methods

The difference in perception was evaluated by changing the variables (Duration, Overlap and Play Speed) of the carrier and modulating wave. The scale of the Duration variable of this experiment was the same as that of experiment 1. Then we set 3 levels for the overlapping state of the modulating wave, namely **State1** (SOA equals Duration), **State2** (use of SOA formula [Israr and Poupyrev 2011]) and **State3** (SOA and decay time equal Duration). Finally, the Play Speed of audio was set with two levels, 1.0x and 2.0x. All

combinations of variables were shown in Supplemental File 1. Each participant received 54 trials ($3\text{Duration} \times 3\text{Overlap} \times 2\text{Play Speed} \times 3\text{repetitions}$) which were divided into 3 sessions. In each trial, we would use 3 different Likert Scales to measure the perception scales of Continuous Motion, Speed of Transmission and Satisfaction, and performed a repeated measures ANOVA to analyze Continuous Motion, Speed of Transmission and Satisfaction. (Supplemental File 2):

5.2 Results of Experiment 2

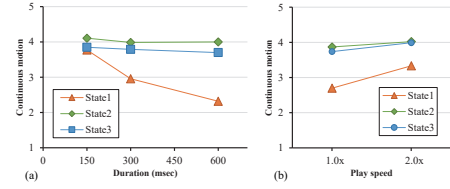


Figure 5: Average continuous motion of overlap state at different (a) duration and (b) play speed levels.

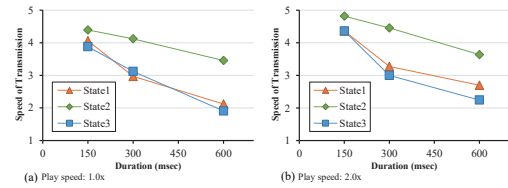


Figure 6: (a) and (b) are the average speed of Transmission for overlap state at different durations and play speeds.

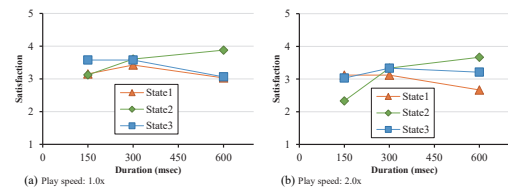


Figure 7: (a) and (b) are the average satisfaction of overlap state at different durations and play speeds.

5.3 Discussion

The ANOVA results of Continuous Motion showed the main effect of Duration, Overlap and Play Speed, and the Duration-Overlap interaction was significant. The comparison of **State2** [11] state with experiment 1 could prove that the proportion of the overlap to DOS had a significant impact on the sense of continuous motion. When the ratio was within a specific interval, the sense of continuous motion was high. Otherwise, it might result in fuzzy or stuck feeling. The conclusion of experiment 1 about recognition rate (Figure 4a) could explain that the high sense of continuous motion (**Dur150**) generated under blur feeling was just an illusion of perception. To improve the play speed of audio could also enhance the participants' sense of speed. The phenomenon might be explained with

synesthesia [19], that is, the change in cognition on the content of the audio signals affects the perception of continuity and speed of the tactile signals. But at the same time, the acceleration of the play speed of audio will also cause the water flow signals to be distorted, which makes the flow felt unnatural to result in low satisfaction.

Finally, it can be deduced that the length of the DOS is supposed to match the rhythm and content of the audio signals.

6 EXPERIMENT 3: UNDERWATER STEERING STATE OF THE TACTILE STIMULATION

This experiment had three purposes. The first was to study whether further changes on the modulating wave could achieve the function of steering. The second was to evaluate the strength of tactile stimulus. The third was to evaluate the satisfaction with the tactile experience.

6.1 Methods

We evaluated the difference in perception by further changing the three variables (Direction, Duration and Function) of the modulating wave. The scale of the Direction variable in this experiment was as shown in Figure 8 with two levels of **R2L** (R1 to L3 transmission) and **L2R** (L1 to R3 transmission). The scale of the Duration variable was also the same as experiment 1. There are 3 steering Functions corresponding to the Direction, namely **Multiply** (one-sided DOS is twice that of the other side), **Delay** (one-sided stimulus onset is delayed by 1/3 times the DOS compared to the other side) and **Volume** (one-sided amplitude decreases, the other side amplitude increases) functions. All combinations of variables were shown in Supplemental File 1. In each trial, we would use single-choice questions to measure Success Rate, and the rest would use two different Likert Scales to measure the sensing scales of Signal Strength and Satisfaction, and performed a repeated measures ANOVA to analyze Success Rate, Signal Strength and Satisfaction (Supplemental File 2):

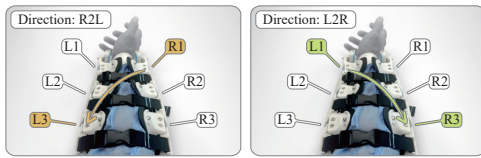


Figure 8: Two variables of direction.

6.2 Result of Experiment 3

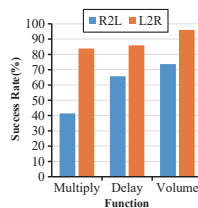


Figure 9: The success rate of two directions at different functions.

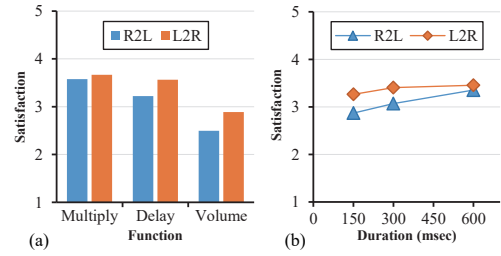


Figure 10: Average satisfaction of two directions at different (a) function and (b) duration levels.

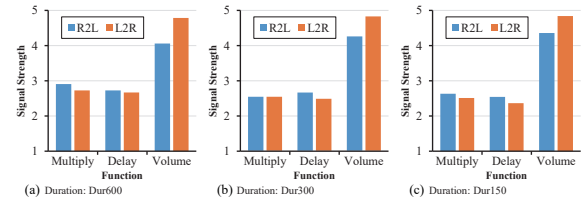


Figure 11: (a), (b) and (c) are the average signal strength of direction at different durations and functions.

6.3 Discussion

It could be seen that the recognition rate of steering state in the **L2R** direction was significantly higher than that in the **R2L** direction. The main reason for this phenomenon could be presumed to be the difference in tactile sensitivity [18] between the inner and outer sides of arms. The success rate of steering state of the **Volume** and **Delay** functions was significantly higher than that of the **Multiply** function. Therefore, it was speculated that the difference in the onset of the tactile signals on both sides would provide a suggestion for the direction to increase the participants' success rate of steering state. As for the **Volume** function alone, the control of the amplitude of the modulating wave resulted in an intensity which made the participants have higher recognition rate of steering state but lower evaluation of this function. This reflected that the tactile stimulus was suitable to resemble the sense of strike rather than flow because of high intensity.

7 CONCLUSION

This research results further prove the argument with the conventional audio tools as the haptic framework in the past [12]. It was confirmed in experiment 1 that the perception of tactile texture and tactile motion could exist simultaneously and be identified. The results of experiment 2 illustrated the possible control parameters of the sense of flow, which further indicated the high correlation between audio parameters (e.g., rhythm, timbre, tempo, loudness, etc.) and tactile experience. It was proved in experiment 3 that people could feel the difference between the tactile signals on both sides and the information contained in them. When tactile experience is combined with technologies such as Virtual Reality and Augmented Reality, the design of the information carried by the tactile signal requires further consideration of people's existing cognition and subjective feelings.

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